

A Quantum leap for computing

"Where... the ENIAC is equipped with 18,000 vacuum tubes and weighs 30 tons, computers in the future may have 1,000 vacuum tubes and perhaps weigh just 1-1/2 tons." Popular Mechanics, March 1949

Things have changed. Things will change again. The world we see around us, is an average of possibilities. There are laws that govern its operation, and are hard to comprehend. Newton did not discover them, Einstein did not like them. Doesn't matter. Quantum Mechanics describes all physical structure, and is the only set of laws that apply to the extremely small particles that make the universe. It allows for strange behaviour, and works on probability. It opens the doors to computing power that we cannot yet fully comprehend. Quantum computers harness the sub-atomic particles that form the fabric of the universe.

Computing with physics

Quantum technologies can be used in two very different ways to create computers that perform far better than the best current technologies can hope to achieve. One method, called Quantum Cellular Automata (QCA) is similar to